Mid-term Exam: Statistical Inference

1 (20') Let $X_1, \cdots, X_n$ be a random sample from $N(\mu_0, \sigma^2)$, where $\mu_0 > 0$ is known and $\sigma > 0$ is unknown.

(i) (5') Derive the sufficient and complete statistic of σ^2;

(ii) (5') Derive the MLE and UMVUE of $\sigma + \sigma^3$;

(iii) (5') Derive the C-R lower bound for the unbiased estimator of σ^4;

(iv) (5') Is the MLE of σ^4 an asymptotically effective estimator of σ^4?

2 (20') Let $X_1, \cdots, X_n$ be a random sample from a population with distribution function

$$F(x) = \begin{cases} 1 - \exp\left\{-\frac{x-\alpha}{\beta}\right\}, & x \geq \alpha, \\ 0, & x < \alpha, \end{cases} \qquad -\infty < \alpha < \infty, \quad \beta > 0.$$

(i) (4') Is this family an exponential family?

(ii) (5') When α is known, derive the sufficient statistic and MLE of β;

(iii) (5') When β is known, derive the sufficient statistic and MLE of α;

(iv) (6') When both α and β are unknown, derive the sufficient statistic and MLE of (α, β).

3 (15') Let $X_1, \cdots, X_m$ and $Y_1, \cdots, Y_n$ be independent random samples from Normal distributions $N(\theta, \sigma_1^2)$ and $N(\theta, \sigma_2^2)$, respectively, where θ is unknown, σ_1^2 and σ_2^2 are known. For any constant c with $0 \leq c \leq 1$, set $U_c = c\bar{X} + (1-c)\bar{Y}$, where $\bar{X}$ and $\bar{Y}$ are the sample means of the X_i's and Y_i's,

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(i) (5') Show that U_c is an unbiased estimate of θ for every c;

(ii) (5') Calculate the variance of U_c, and show that it is minimized at $c = c_0 = m\sigma_2^2/(n\sigma_1^2 + m\sigma_2^2)$;

(iii) (5') Show that U_{c_0} for the above defined c_0 is the UMVUE of θ.

4 (15') Suppose that $X_1, \cdots, X_m$ i.i.d. $\sim N(\mu_1, \sigma^2)$, $Y_1, \cdots, Y_n$ i.i.d. $\sim N(\mu_2, \sigma^2)$, where μ_1, μ_2, σ^2 are unknown, and X_i's and Y_j's are independent. Let $\bar{X}, \bar{Y}, S_X^2, S_Y^2$ denote their sample means and sample variances.

(i) (10') Derive the MLE and UMVUE of $\mu_1 - \mu_2$ and σ^2;

(ii) (5') For fixed constants α and β, determine the distribution of $T = \{\alpha(\bar{X} - \mu_1) - \beta(\bar{Y} - \mu_2)\}/\sqrt{(m-1)S_X^2 + (n-1)S_Y^2}$.

5 (30') Suppose that the observed data (y_i, x_i) follows a linear regression model without an intercept, $y_i = \beta x_i + \epsilon_i$, $i = 1, \cdots, n$, where $x_i \in R$ is fixed (not random) and known, β is an unknown one-dimensional parameter, and $\epsilon_i, i = 1, \cdots, n$ are i.i.d. $\sim N(0, \sigma^2)$ with σ^2 is unknown.

(i) (10') Derive the MLEs of β and σ^2;

(ii) (10') Denote the MLEs of β and σ^2 as $\hat{\beta}$ and $\hat{\sigma}^2$. Determine the distribution of $\hat{\beta}$ and $\hat{\sigma}^2$;

(iii) (5') Is $\hat{\beta}$ independent of $\hat{\sigma}^2$? Why?

(iii) (5') Show that U_{c_0} for the above defined c_0 is the UMVUE of θ.

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(ii) (5') For fixed constants α and β, determine the distribution of $T = \{\alpha(\bar{X} - \mu_1) - \beta(\bar{Y} - \mu_2)\}/\sqrt{(m-1)S_X^2 + (n-1)S_Y^2}$.

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(i) (10') Derive the MLEs of β and σ^2;

(ii) (10') Denote the MLEs of β and σ^2 as $\hat{\beta}$ and $\hat{\sigma}^2$. Determine the distribution of $\hat{\beta}$ and $\hat{\sigma}^2$;

(iii) (5') Is $\hat{\beta}$ independent of $\hat{\sigma}^2$? Why?

(iv) (5') The observed values of (y_i, x_i) are given in the following table. Calculate the observed values of $\hat{\beta}$ and $\hat{\sigma}^2$.

x_i	5	10	15	20	25	30
y_i	0.10	0.21	0.30	0.35	0.41	0.52